| Question | | | Marking details | Marks available | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | AO1 | AO2 | AO3 | Total | Maths | Prac |
| **1** | (a) | (i) | [The *activity* is the] rate of decay or [the number of] disintegrations / decays per second (or unit time [of a sample of radioactive nuclei] (1) Accept number of radioactive emissions per second<br>Becquerel or becquerel or Bq (1) Accept $s^{-1}$ | 2 | | | 2 | | |
| | | (ii) | $\lambda = \frac{\ln 2}{3.3 \times 10^5} = 2.1 \times 10^{-6}\ \text{s}^{-1}$ (1)<br>$e^{\lambda t} = \frac{A_0}{A}$;  $t = \frac{1}{\lambda}\ln\left(\frac{A_0}{A}\right)$ (1)<br>$t = \frac{1}{(2.1\times10^{-6})}\ln\left(\frac{1}{0.2}\right) = 7.7 \times 10^5\ \text{s} = [8.9\ \text{day}]$ (1)<br><br>**Alternative:**<br>Taking logs of $A = \frac{A_0}{2^n}$ e.g. $\ln A = \ln A_0 - n\ln 2$ or $n\ln 2 = \ln\frac{A_0}{A}$ (1)<br>Substitution: $n\ \ln 2 = \ln\frac{100}{20}$ [leads to $n$ = 2.32] (1)<br>Correct answer $7.7 \times 10^5$ s = [8.9 day] (1) | | 3 | | 3 | 3 | |
| | | (iii) | Safe outside the body or stopped by the skin or danger or dangerous if swallowed / inhaled (1)<br><u>Highly</u> ionising (1)<br>Any reference to damage e.g. damage DNA, kills cells, cause cancer, cause radiation poisoning, causes mutations (1) | | | 3 | 3 | | |